

POWER MANAGEMENT

The schematic diagram illustrates the power management circuit for the RAK4630 module. It is divided into two main sections by a vertical dashed line.

Left Section (J1 Connector):

- J1 Conn_01x03:** A 3-pin header connector.
- BATT_NTC:** A red box representing the battery NTC sensor, connected to pin 3 of J1.
- VBAT_RAW+:** A red box representing the raw battery voltage input, connected to pin 1 of J1.
- GND:** Ground connection for the BATT_NTC and VBAT_RAW+ pins.
- DMG2305UX-13 (D2):** A yellow box representing the DC-DC converter. Its pin 1 is connected to GND, pin 2 is connected to VBAT_RAW+, and pin 3 is connected to the VBAT line.

Right Section (VBAT and Voltage Divider):

- VBAT:** The output of the DC-DC converter, which also serves as the input to the voltage divider.
- Voltage Divider:** Consists of two resistors, R6 (1M) and R7 (1M), connected in series between VBAT and GND.
- C3 (100nF):** A capacitor connected in parallel with the voltage divider resistors to filter the signal.
- VBAT_SENSE:** A red box representing the sensed battery voltage, connected to the node between R6 and R7.
- Notes:**
 - Voltage divider 2:1
 - For RAK4630 AIN2/P0.04
 - Backup Voltage measurement

takes 5v from USB, outputs to:
Pin:2VBAT>DMG230_Diode>Pin2:VBAT_RAW (to charge the battery)
Pin10:VOUT_SYS > TPS6284_BUCK Pin2: VIN

VBUS_5V

C2 1uF

R4 10k

GND

VCC_3v3

R2 1k

GND

R5 1k

GND

GND

U2 BQ24074RGT

ITEM 15

OUT 10

VBAT 2

TS 1

PGOOD 7

CHG 9

VS 8

ISET 16

ILIM 12

EN2 5

EN1 6

TMR 14

CE 4

VOUT_SYS

BATT_NTC

LiPo Battery charger IC

X

X

GND

3v3 Buck Regulator
Input: VOUT_SYS from BQ24074 OUT
Converts to stable/constant VCC_3.3v for the system

U3
TPS62840DLCR

VIN SW
EN VOS
MODE
STOP
VSET
GND

C1
C 10uF

C4
10uF

C5
10uF

L1
2.2uH

PWR_FLAG

VCC_3v3

GND

The schematic shows the MAX17048G_T10 chip, labeled 'Fuel guage' and 'Battery level monitor'. It has several pins: CTG (1), CELL (2), VDD (3), EP (9), QSTRT (6), SCL (7), SDA (8), ALERT (5), and GND (4). The chip is connected to a 3V3 supply (VCC_3v3) via a 4.7k resistor (R3). The ALERT pin is connected to a BATT_ALERT output. The PWR_FLAG pin is connected to a VBAT_RAW+ input. The SDA and SCL pins are connected to a BATT_I2C_SDA and BATT_I2C_SCL input. The chip is also connected to a 100nF capacitor (C6) and a GND pin.

single wire debug (SWD) socket

VCC_3v3
SWDIO
NRF_RESET
SWDCLK
GND

J6
Conn_01x06

UART_2 to GM02sp

VCC_3v3

UART_TX

UART_RX

GND

J3
Conn_01x04

The diagram shows a circuit with a green wire running vertically. A 10k resistor, labeled R16, is connected to a VCC_3V3 supply. A 1uF capacitor, labeled C8, is connected to ground.

Diagram showing two pins labeled `VCC_3v3` connected to a common ground rail.

The circuit diagram illustrates a 4-bit parallel adder implemented using two 74LS148 8-to-3 priority encoders (SW1 and SW2) and two 74LS148 8-to-3 priority decoders (D1, D3, D4, D5). The two 74LS148 encoders are configured to take 4-bit inputs (A, B, C, D) and produce 3-bit outputs (S, C, R). The two 74LS148 decoders are configured to take the 3-bit outputs (S, C, R) and produce 4-bit outputs (S, C, R, D). The circuit is powered by a 5V supply and includes pull-up resistors (10k) for the inputs and outputs. The ground connections are labeled GND.

A diagram showing a horizontal line with a vertical line intersecting it. The horizontal line is labeled '3' on the left and '4' on the right. The vertical line is labeled 'GND' at the top.

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RAK4630(NRF52840 MCU + SX1280)



PS_STATUS
RESET
GND
UART_TX
UART_RX

Pin connection diagram for the USB module. The diagram shows a 13-pin connector on the left with pins labeled 9/NFC1, 10/NFC2, 11/RT1_DE, 12/RT1_RX, 13/RT2_DE, 14/RT2_RX, 15/RT2_TX, 16/RT2_SCL, 17/RT2_SDA, 18/USB+, 19/USB-, and 20/VBUS. On the right, various components are connected: VCC_3v3 to pins 9 and 10; Conn_01x02_Pin to pins 11 and 12; BATT-ALERT to pin 13; WAKE to pin 14; BATT_I2C_ to pins 15 and 16; BATT_I2C_ to pins 17 and 18; USB_D+ to pin 19; USB_D- to pin 20; and VBUS to pin 20. A note indicates that pins 19 and 20 are for 5V power supply.

[illegible]

Pin	Signal
1	P0.26/QSP_L
2	P0.30/QSP_L
3	P0.28/QSP_L
4	P0.02/QSP_L
5	P0.03/QSP_L
23	P0.24/I2C_S
24	P0.25/I2C_S
25	P1.01/SW1
26	P1.02/SW2
27	P1.03/LED1
28	P1.04/LED2
29	P0.03/QSP_L
30	P0.02/QSP_L
31	P0.28/QSP_L
32	P0.26/QSP_L
33	P0.30/QSP_L
34	P0.03/QSP_L
35	GND_4

The diagram shows a 4-bit parallel adder circuit. A 74181 ALU is configured for addition with carry-in 1. The 74148 priority encoder provides carry-in signals to the ALU inputs. The ALU outputs are connected to a 7-segment display.

74181 ALU Configuration:

- Inputs:** A, B, C, D (4-bit parallel inputs).
- Carry-in:** 1 (connected to the carry-in input).
- Outputs:** F₁, F₂, F₃, F₄ (4-bit parallel outputs).

74148 Priority Encoder Configuration:

- Inputs:** I₁, I₂, I₃, I₄, I₅, I₆, I₇ (7-bit parallel inputs).
- Outputs:** O₁, O₂, O₃, O₄ (4-bit parallel outputs).

7-segment Display:

- Inputs:** A, B, C, D (4-bit parallel inputs).
- Outputs:** F₁, F₂, F₃, F₄ (4-bit parallel outputs).

Diagram illustrating a PCB layout with a GND connection and a table of project information.

GND Connection: A red arrow points down to a green pad labeled "GND".

Table:

Sheet:	
File:	
Title:	
Size:	
KiCad:	

7

8

MH1
M2_Screw

MH2
M2_Screw

(1262 Lora RFIC)

Diagram showing pin assignments for the GM02SP module:

Pin	Signal
4	UART_TX
5	UART_RX
6	GM02SP_PS_STATUS
7	WAKE
8	GM02SP_RESET

Shield		GND	
Power rails: -V_BUS: 5v from USB charging -VBAT: Raw battery from LiPo Cell 3.0v (min) to 3.7v(typ) to 4.2v(Max) -VCL_5V0: regulated 3.3 volts -v4_OUT: 1.8v output from GMD25P to simcard & peripherals			
BPV4_Collar.kicad_sch			
Rev:			
A3		Date:	
d E.D.A. 9.0.7		Id: 1/1	

TODO: -check jst polarity	Power rails: -V_BUS: 5v from USB charging -VBAT: Raw battery from Lipo Cell 3.0v (min) to 3.7v(typ) to 4.2v(Max) -VCC_3v3: regulated 3.3 volts -lv6_OUT: 1.8v output from GM025P to simcard & peripherals
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Sheet: /
File: BPV4_Collar.kicad_sch

Title:		
Size: A3	Date:	Rev:

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 1/1

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 1/1